SWORD v17c Variable Reference

Quick-lookup for all variables in the v17c NetCDF export files (`{region}_sword_v17c_beta.nc`).

Fill values: i4 = `-9999`, i8 = `-9999`, f8 = `-9999.0`

Distance Variable Conventions

SWORD contains four distance-to-outlet variables. They differ in routing method, zero-point, and measurement anchor within a reach. The table below summarizes these conventions; see "Node-Level Interpolation" for how each extends to nodes.

Variable	Routing	Reach-level anchor	Outlet value	Ghost reaches
`dist_out` (v17b)	v17b topology	upstream end of reach	`reach_length`	has value
`hydro_dist_out`	`rch_id_dn_main` chain	upstream end of reach	`reach_length`	has value
`dist_out_dijkstra`	Dijkstra shortest path	upstream end of reach	**0**	NULL, except isolated ghost coastal outlets use `reach_length`
`hydro_dist_hw`	`rch_id_up_main` chain	upstream end of reach	0	has value

Anchor convention. All four distance variables anchor at the upstream end of the reach. For `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra`, the value equals the distance from the upstream end of the reach to the outlet; each reach exceeds its downstream neighbor by its own `reach_length`. For `hydro_dist_hw`, the value equals the distance from the headwater to the upstream end of the reach; each reach exceeds its upstream neighbor by the upstream neighbor's `reach_length`.

Zero-point difference. `dist_out` and `hydro_dist_out` assign the outlet reach a value equal to its `reach_length` (the upstream end is one reach_length from the outlet point). `dist_out_dijkstra` assigns the outlet reach a value of 0 (the outlet point itself is the reference). When comparing these variables, the offset at any reach equals the outlet's `reach_length`.

Ghost reaches. `dist_out_dijkstra` is NULL for ghost reaches (type=6) except isolated ghost coastal outlets (`type=6`, `end_reach=2`, no downstream neighbor), which use `reach_length` at reach level and interpolate from that value at nodes. The other three variables retain values for ghost reaches.

Node-Level Interpolation

Six distance variables are interpolated per node using a midpoint offset within the parent reach: `offset = cumsum(node_length) - 0.5 * node_length`, where the cumulative sum is ordered by `node_order`. This places each node at the geometric center of its `node_length` segment. On a single-path network (no junctions), `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra` are exactly equal at every node.

Variable	Node formula	node_order=n (upstream)	node_order=1 (downstream)
`dist_out`	`reach.do - reach_length + offset`	`~ reach.do`	`~ reach.do - reach_length`
`hydro_dist_out`	`reach.hdo - reach_length + offset`	`~ reach.hdo`	`~ reach.hdo - reach_length`
`dist_out_dijkstra`	`reach.dod - reach_length + offset`	`~ reach.dod`	`~ reach.dod - reach_length`
`hydro_dist_hw`	`reach.hdh + reach_length - offset`	`~ reach.hdh`	`~ reach.hdh + reach_length`
`pathlen_hw`	`reach.plh - reach_length + offset`	`~ reach.plh - reach_length`	`~ reach.plh`
`pathlen_out`	`reach.plo + reach_length - offset`	`~ reach.plo + reach_length`	`~ reach.plo`

Three additional variables (`subnetwork_id`, `best_headwater`, `best_outlet`) are flat copies from the parent reach.

Boundary continuity. At reach boundaries, the downstream-end node of the upstream reach and the upstream-end node of the downstream reach share identical interpolated distance values (verified gap = 0.00 m across all boundaries).

New v17c Reach Variables

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
dist_out_dijkstra	f8	meters	-9999.0		Dijkstra shortest-path distance from the upstream end of the reach to any network outlet; outlet reach = 0; NULL for ghost reaches (type=6)
hydro_dist_out	f8	meters	-9999.0		Mainstem distance from the upstream end of the reach to best_outlet via rch_id_dn_main chain; outlet reach = reach_length
hydro_dist_hw	f8	meters	-9999.0		Mainstem distance from best_headwater to the upstream end of the reach via rch_id_up_main chain; headwater = 0

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
rch_id_up_main	i8		-9999		Main upstream neighbor reach ID (mainstem-preferred)
rch_id_dn_main	i8		-9999		Main downstream neighbor reach ID (mainstem-preferred)
subnetwork_id	i4		-9999		Connected component ID (Pfafstetter-offset, globally unique; differs from v17b `network`)
main_path_id	i8		-9999		ID of the mainstem path this reach belongs to
is_mainstem	i4		-9999	flag_values=[0,1], flag_meanings="not_mainstem mainstem"	Whether reach is on a mainstem path
best_headwater	i8		-9999		Width-prioritized upstream headwater reach ID

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
best_outlet	i8		-9999		Width-prioritized downstream outlet reach ID
pathlen_hw	f8	meters	-9999.0		Cumulative reach_length sum from best_headwater to the downstream end of the reach (headwater = 0, increases downstream)
pathlen_out	f8	meters	-9999.0		Cumulative reach_length sum from the upstream end of the reach to best_outlet (outlet = 0, increases upstream)
facc_quality	i4		-9999	flag_values=[1], flag_meanings="denoise_v3"	Flow accumulation correction flag
dl_grod_id	i8		-9999		Dam/lake GROD ID (downstream lookup)
wse_obs_p10	f8	meters	-9999.0		SWOT WSE 10th percentile

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
wse_obs_p20	f8	meters	-9999.0		SWOT WSE 20th percentile
wse_obs_p30	f8	meters	-9999.0		SWOT WSE 30th percentile
wse_obs_p40	f8	meters	-9999.0		SWOT WSE 40th percentile
wse_obs_p50	f8	meters	-9999.0		SWOT WSE 50th percentile (median)
wse_obs_p60	f8	meters	-9999.0		SWOT WSE 60th percentile
wse_obs_p70	f8	meters	-9999.0		SWOT WSE 70th percentile
wse_obs_p80	f8	meters	-9999.0		SWOT WSE 80th percentile
wse_obs_p90	f8	meters	-9999.0		SWOT WSE 90th percentile
wse_obs_range	f8	meters	-9999.0		SWOT WSE range (p90 - p10)
wse_obs_mad	f8	meters	-9999.0		SWOT WSE median absolute deviation
width_obs_p10	f8	meters	-9999.0		SWOT width 10th percentile
width_obs_p20	f8	meters	-9999.0		SWOT width 20th percentile
width_obs_p30	f8	meters	-9999.0		SWOT width 30th percentile

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
width_obs_p40	f8	meters	-9999.0		SWOT width 40th percentile
width_obs_p50	f8	meters	-9999.0		SWOT width 50th percentile (median)
width_obs_p60	f8	meters	-9999.0		SWOT width 60th percentile
width_obs_p70	f8	meters	-9999.0		SWOT width 70th percentile
width_obs_p80	f8	meters	-9999.0		SWOT width 80th percentile
width_obs_p90	f8	meters	-9999.0		SWOT width 90th percentile
width_obs_range	f8	meters	-9999.0		SWOT width range (p90 - p10)
width_obs_mad	f8	meters	-9999.0		SWOT width median absolute deviation
slope_obs_p10	f8	meters/meters	-9999.0		SWOT slope 10th percentile
slope_obs_p20	f8	meters/meters	-9999.0		SWOT slope 20th percentile
slope_obs_p30	f8	meters/meters	-9999.0		SWOT slope 30th percentile
slope_obs_p40	f8	meters/meters	-9999.0		SWOT slope 40th percentile

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
slope_obs_p50	f8	meters/meters	-9999.0		SWOT slope 50th percentile (median)
slope_obs_p60	f8	meters/meters	-9999.0		SWOT slope 60th percentile
slope_obs_p70	f8	meters/meters	-9999.0		SWOT slope 70th percentile
slope_obs_p80	f8	meters/meters	-9999.0		SWOT slope 80th percentile
slope_obs_p90	f8	meters/meters	-9999.0		SWOT slope 90th percentile
slope_obs_range	f8	meters/meters	-9999.0		SWOT slope range (p90 - p10)
slope_obs_mad	f8	meters/meters	-9999.0		SWOT slope median absolute deviation
slope_obs_adj	f8	meters/meters	-9999.0		Adjusted SWOT slope (bias-corrected)
slope_obs_slopeF	f8		-9999.0		Slope quality F-statistic
slope_obs_reliable	i4		-9999	flag_values=[0,1], flag_meanings="unreliable reliable"	Whether SWOT slope is reliable

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
slope_obs_quality	i4		-9999	flag_values=[0..8], flag_meanings="reliable small_negative moderate_negative large_negative negative below_ref_uncertainty high_uncertainty noise_high_nobs flat_water_noise"	SWOT slope quality category
slope_obs_n	i8		-9999		Number of RiverSP node observations used in pass-level slope fits
slope_obs_n_passes	i8		-9999		Number of SWOT passes with slope
slope_obs_q	i8		-9999	Integer bitfield (1=negative, 2=low_passes, 4=high_var, 8=extreme, 16=clipped)	SWOT slope quality bitfield
n_obs	i4		-9999		Total number of SWOT observations

New v17c Node Variables

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
subnetwork_id	i4		-9999		Connected component ID (from parent reach)
best_headwater	i8		-9999		Width-prioritized upstream headwater reach ID
best_outlet	i8		-9999		Width-prioritized downstream outlet reach ID
pathlen_hw	f8	meters	-9999.0		Cumulative path length from headwater, interpolated by node position within reach; single-node reaches use midpoint

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
pathlen_out	f8	meters	-9999.0		Cumulative path length to outlet, interpolated by node position within reach; single-node reaches use midpoint
hydro_dist_out	f8	meters	-9999.0		Mainstem distance to best_outlet, interpolated by node position within reach; single-node reaches use midpoint
hydro_dist_hw	f8	meters	-9999.0		Distance from best_headwater, interpolated by node position within reach; single-node reaches use midpoint

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
dist_out_dijkstra	f8	meters	-9999.0		Dijkstra shortest-path distance to outlet, interpolated by node position within reach; single-node reaches use midpoint; NULL for ghost reaches
wse_obs_p10	f8	meters	-9999.0		SWOT WSE 10th percentile
wse_obs_p20	f8	meters	-9999.0		SWOT WSE 20th percentile
wse_obs_p30	f8	meters	-9999.0		SWOT WSE 30th percentile
wse_obs_p40	f8	meters	-9999.0		SWOT WSE 40th percentile
wse_obs_p50	f8	meters	-9999.0		SWOT WSE 50th percentile (median)
wse_obs_p60	f8	meters	-9999.0		SWOT WSE 60th percentile
wse_obs_p70	f8	meters	-9999.0		SWOT WSE 70th percentile
wse_obs_p80	f8	meters	-9999.0		SWOT WSE 80th percentile

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
wse_obs_p90	f8	meters	-9999.0		SWOT WSE 90th percentile
wse_obs_range	f8	meters	-9999.0		SWOT WSE range (p90 - p10)
wse_obs_mad	f8	meters	-9999.0		SWOT WSE median absolute deviation
width_obs_p10	f8	meters	-9999.0		SWOT width 10th percentile
width_obs_p20	f8	meters	-9999.0		SWOT width 20th percentile
width_obs_p30	f8	meters	-9999.0		SWOT width 30th percentile
width_obs_p40	f8	meters	-9999.0		SWOT width 40th percentile
width_obs_p50	f8	meters	-9999.0		SWOT width 50th percentile (median)
width_obs_p60	f8	meters	-9999.0		SWOT width 60th percentile
width_obs_p70	f8	meters	-9999.0		SWOT width 70th percentile
width_obs_p80	f8	meters	-9999.0		SWOT width 80th percentile
width_obs_p90	f8	meters	-9999.0		SWOT width 90th percentile

Variable	NetCDF Type	Units	Fill Value	Encoding	Description
width_obs_range	f8	meters	-9999.0		SWOT width range (p90 - p10)
width_obs_mad	f8	meters	-9999.0		SWOT width median absolute deviation
n_obs	i4		-9999		Total number of SWOT observations
facc_quality	i4		-9999	flag_values=[1], flag_meanings="denoise_v3"	Flow accumulation correction flag

v17b Variables (unchanged)

Reaches

Variable	NetCDF Type	Units	Fill Value	Description
reach_id	i8		-9999	Reach identifier (format: CBBBBBRRRRT)
x	f8	degrees east	-9999.0	Reach centroid longitude
x_min	f8	degrees east	-9999.0	Minimum longitude of reach extent
x_max	f8	degrees east	-9999.0	Maximum longitude of reach extent
y	f8	degrees north	-9999.0	Reach centroid latitude

Variable	NetCDF Type	Units	Fill Value	Description
y_min	f8	degrees north	-9999.0	Minimum latitude of reach extent
y_max	f8	degrees north	-9999.0	Maximum latitude of reach extent
reach_length	f8	meters	-9999.0	Length of reach centerline
n_nodes	i4		-9999	Number of nodes in reach
wse	f8	meters	-9999.0	Water surface elevation (MERIT Hydro)
wse_var	f8	meters^2	-9999.0	WSE variance
width	f8	meters	-9999.0	River width (GRWL)
width_var	f8	meters^2	-9999.0	Width variance
facc	f8	km^2	-9999.0	Flow accumulation (MERIT Hydro)
n_chan_max	i4		-9999	Max number of channels at any node
n_chan_mod	i4		-9999	Modal number of channels
obstr_type	i4		-9999	Obstruction type (0=none, 1=dam, 2=lock, 3=low-head, 4=falls)
grod_id	i8		-9999	GRanD/GROD dam ID
hfalls_id	i8		-9999	High falls ID

Variable	NetCDF Type	Units	Fill Value	Description
slope	f8	meters/kilometers	-9999.0	Water surface slope from MERIT DEM
dist_out	f8	meters	-9999.0	Distance from upstream end of reach to network outlet (v17b original; outlet reach = reach_length, not 0). See “Distance Variable Conventions” for comparison with v17c distance variables
n_rch_up	i4		-9999	Number of upstream neighbor reaches
n_rch_down	i4		-9999	Number of downstream neighbor reaches
lakeflag	i4		-9999	Water body type (0=river, 1=lake, 2=canal, 3=tidal)

Variable	NetCDF Type	Units	Fill Value	Description
type	i8		-9999	Reach type (1=river, 3=lake_on_river, 4=dam, 5=unreliable, 6=ghost). Not in v17b NetCDF; added in v17c. Values from v17b database.
add_flag	i8		-9999	Manually added reach flag
swot_obs	i4		-9999	Number of expected SWOT observations per cycle
swot_obs_source	string			Source of SWOT observation data
max_width	f8	meters	-9999.0	Maximum width at any node
low_slope_flag	i4		-9999	Low slope flag
trib_flag	i4		-9999	MHV tributary enters (0=no, 1=yes, spatial proximity)
path_freq	i8		-9999	Traversal count (increases toward outlets)
path_order	i8		-9999	Path order from traversal

Variable	NetCDF Type	Units	Fill Value	Description
path_segs	i8		-9999	Unique ID for (path_order, path_freq) combination
stream_order	i4		-9999	Stream order: round(log(path_freq)) + 1
main_side	i4		-9999	Channel role (0=main, 1=side, 2=secondary outlet)
end_reach	i4		-9999	Endpoint type (0=middle, 1=headwater, 2=outlet, 3=junction)
network	i4		-9999	Connected component ID
dn_node_id	i8		-9999	Downstream boundary node ID
up_node_id	i8		-9999	Upstream boundary node ID
river_name	string			River name from GRWL
river_name_en	string			River name (English)
river_name_local	string			River name (local language)
edit_flag	string			Edit provenance tag (e.g. lake_sandwich, harp_lake)
version	string			Data version identifier

Nodes

Variable	NetCDF Type	Units	Fill Value	Description
node_id	i8		-9999	Node identifier (format: CBBBBBR-RRRNNNT)
x	f8	degrees east	-9999.0	Node longitude
y	f8	degrees north	-9999.0	Node latitude
node_length	f8	meters	-9999.0	Length of river represented by this node
node_order	i4		-9999	1-based position within reach (1=down-stream, n=upstream, by dist_out)
reach_id	i8		-9999	Parent reach ID (format: CBBBBBR-RRRT)

`0.0.5` release note (April 2026): the corrected files supersede the prior `0.0.4` beta uploads. Published NetCDF, GeoPackage, and Parquet files were reissued to recompute `dn_node_id`, `up_node_id`, and `node_order` directly from node `dist_out`, restore ghost coastal outlet `dist_out_dijkstra`, and align single-node `node.dist_out` with the same midpoint anchor used by the other node-level outlet distances. | wse | f8 | meters | -9999.0 | Water surface elevation (MERIT Hydro) | | wse_var | f8 | meters^2 | -9999.0 | WSE variance | | width | f8 | meters | -9999.0 | River width (GRWL) | | width_var | f8 | meters^2 | -9999.0 | Width variance | | n_chan_max | i4 | | -9999 | Max number of channels | | n_chan_mod | i4 | | -9999 | Modal number of channels | | obstr_type | i4 | | -9999 | Obstruction type | | grod_id | i8 | | -9999 | GRanD/GROD dam ID | | hfalls_id | i8 | | -9999 | High falls ID | | dist_out | f8 | meters | -9999.0 | Distance to network outlet. Interpolated from reach `dist_out` using node_length midpoint offsets (same formula as `hydro_dist_out` and `dist_out_dijkstra`). On single-path networks all three are exactly equal | | wth_coef | f8 | | -9999.0 | Width coefficient | | ext_dist_coef | f8 | | -9999.0 | Extraction distance coefficient | | facc | f8 | km^2 | -9999.0 | Flow accumulation (MERIT Hydro) | | lakeflag | i8 | | -9999 | Water body type (0=river, 1=lake, 2=canal, 3=tidal). In 0.0.11+, propagated from parent reach lakeflag per JPL request (was independently derived from GRWL in prior versions). | | max_width | f8 | meters | -9999.0 | Maximum width | | meander_length | f8 | | -9999.0 | Meander wavelength | | sinuosity | f8 | | -9999.0 | Channel sinuosity | | manual_add | i4 | | -9999 | Manually added node flag | | trib_flag | i4 | | -9999 | MHV tributary enters (0=no, 1=yes) | | path_freq | i8 | | -9999 | Traversal count | | path_order | i8 | | -9999 | Path order | | path_segs | i8 | | -9999 | Path segment ID | | stream_order | i4 | | -9999 | Stream order | | main_side | i4 | | -9999 | Channel role (0=main, 1=side, 2=secondary outlet) | | end_reach | i4 | | -9999 | Endpoint type (0=middle, 1=headwater, 2=outlet, 3=junction) | | network | i4 | | -9999 | Connected component ID | | add_flag | i8 | | -9999 | Manually added node flag | | river_name | string | | | River name | | edit_flag | string | | | Edit provenance tag (e.g. lake_sandwich, harp_lake) | | version | string | | | Data version identifier |

Multi-dimensional Arrays

Variable	Group	Shape	Type	Description
cl_id_min	reaches	[num_reaches]	i8	Minimum centerline ID bounding this reach
cl_id_max	reaches	[num_reaches]	i8	Maximum centerline ID bounding this reach
rch_id_up	reaches	[4, num_reaches]	i8	Up to 4 upstream neighbor reach IDs
rch_id_dn	reaches	[4, num_reaches]	i8	Up to 4 downstream neighbor reach IDs
iceflag	reaches	[366, num_reaches]	i4	Daily ice presence flag (Julian day 1-366)
swot_orbits	reaches	[75, num_reaches]	i8	SWOT orbit IDs that observe this reach

Variable	Group	Shape	Type	Description
cl_id_min	nodes	[num_nodes]	i8	Minimum centerline ID bounding this node
cl_id_max	nodes	[num_nodes]	i8	Maximum centerline ID bounding this node
cl_id	centerlines	[num_points]	i8	Centerline point ID
x	centerlines	[num_points]	f8	Centerline point longitude (degrees east)
y	centerlines	[num_points]	f8	Centerline point latitude (degrees north)
reach_id	centerlines	[num_points]	i8	Parent reach ID for this centerline point
node_id	centerlines	[num_points]	i8	Parent node ID for this centerline point
version	centerlines	[num_points]	string	Data version identifier

Subgroups (copied from v17b)

Subgroup	Parent	Description
reaches/area_fits	reaches	Area-based rating curve fit coefficients
reaches/discharge_models	reaches	Discharge model parameters (organized by constraint/model)